

Fingerprint Enhancement Report

Algorithm: **RHLT**
Input file: 1.png
Generated: 2026-08-31 23:08:02 Malay Peninsula Standard Time

Enhancement Quality Summary

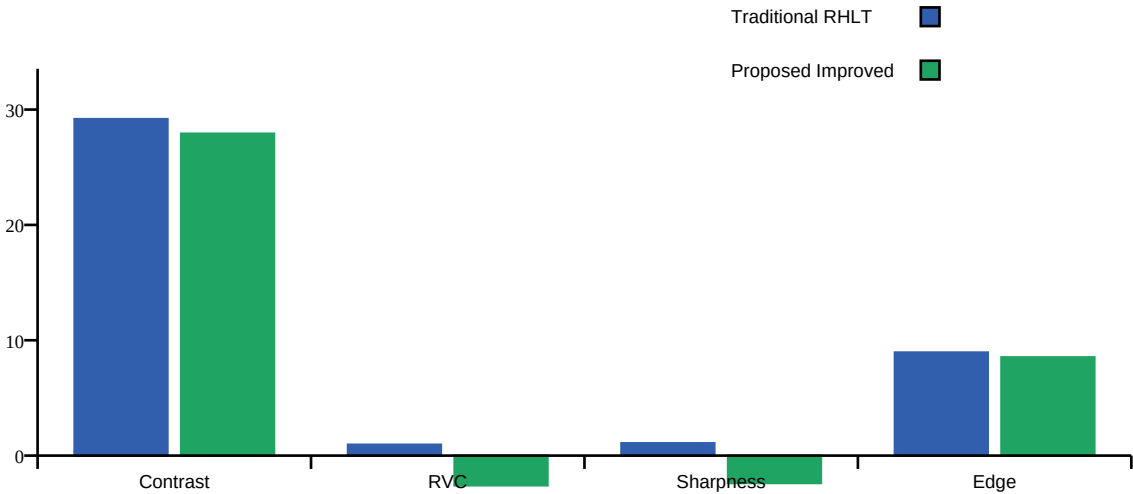
Metric	Value	Interpretation
Contrast Improvement (CII)	1.28x	>= 1.0 = improved
Sharpness Improvement	-2.4%	Positive = sharper ridges
Ridge-Valley Clarity Change	-2.6%	Positive = cleaner ridge/valley separation
SSIM (Structure Preserved)	0.973	>= 0.80 = well preserved
Mean Orientation Coherence	0.613	Closer to 1.0 = more parallel ridge flow
Minutiae Detected	286	Ridge endings + bifurcations on skeleton

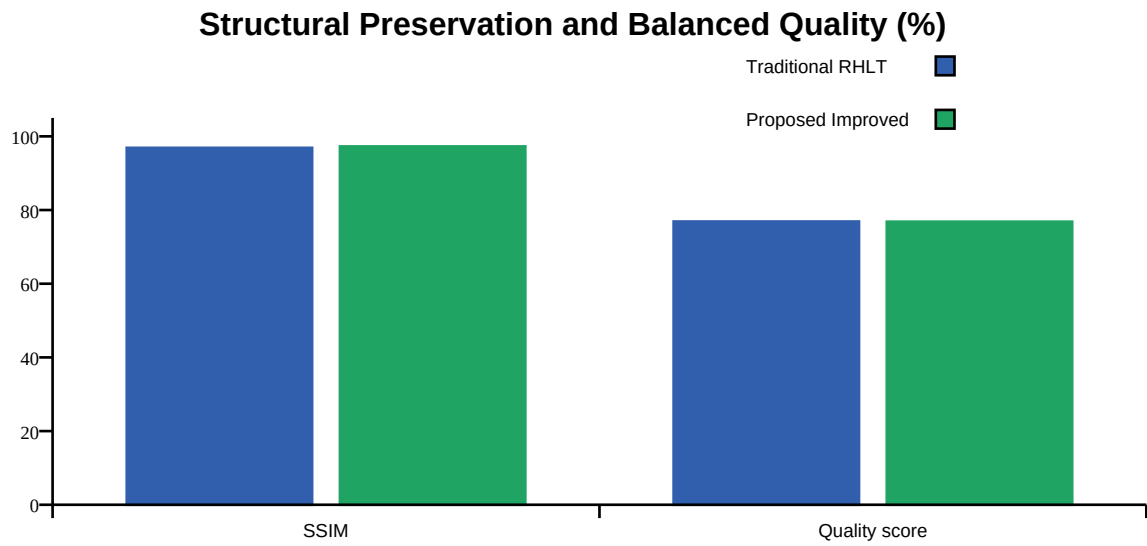
RHLT Candidate Decision

Final selected output: **Proposed Improved RHLT**
Proposed Improved RHLT was selected as the primary method: it passed all structural safety checks and its score difference was -0.0005 (allowed regression tolerance 0.0100).

Metric	Original	Traditional	Proposed	Proposed delta
Contrast	0.1701	0.2196	0.2175	-0.0021
Ridge-Valley Clarity	8,240	8,317	8,029	-288.4
Sharpness (Laplacian)	1,519	1,535	1,483	-52.1
Edge Clarity (Sobel)	57.98	63.16	62.92	-0.24
SSIM (Structural Similarity)	1.000	0.969	0.973	+0.004

Enhancement Change vs Original (%)





Visual Comparison



Original fingerprint



Traditional RHLT Baseline



Proposed Improved RHLT - Final
Selected

Diagnostic Outputs



Ridge orientation field



Ridge binary skeleton



Minutiae overlay

Detailed Metrics: Original vs Enhanced

Metric	Original	Enhanced	Change	Status
Contrast (RMS)	0.1701	0.2175	+0.0474	+
Sharpness (Laplacian)	1,519	1,483	-36.0	-
Edge Clarity (Sobel)	57.98	62.92	+4.93	+
Ridge-Valley Clarity	8,240	8,029	-211.5	-
SSIM (Structure Similarity)	N/A	0.973	N/A	
PSNR (dB)	N/A	22.93	N/A	
MSE	N/A	331.37	N/A	
Foreground Coverage (%)	N/A	17.9	N/A	
Valid Orientation Blocks	N/A	448	N/A	
Mean Orientation Coherence	N/A	0.613	N/A	
Total Minutiae	N/A	286	N/A	
Ridge Endings	N/A	172	N/A	
Bifurcations	N/A	114	N/A	
Processing Time (ms)	N/A	1,800	N/A	